

Huan Wang

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Education

Ph.D.	Yale University , Computer Science	2008–2013
	- Advisor: Prof. Daniel Spielman, Prof. John Wright - Research on clustering, regression, and dictionary learning	
M.Phil.	The Chinese University of Hong Kong , Information Engineering	2005–2007
	- Advisor: Prof. Xiaoou Tang, Prof. Shuicheng Yan, Prof. Jianzhuang Liu - Research on face recognition, manifold learning, subspace learning, semi-supervised learning	
B.Eng.	Zhejiang University , Information Engineering, Chu Kochen Honors College	2000–2004

Professional Experience

Senior Director	Salesforce Research	Palo Alto, CA Feb 2025 – Present
	Large Action Models (xLAM), AI Agents, Reasoning, Planning, SWE Agents, Agent-force	
Director	Salesforce Research	Palo Alto, CA Nov 2021 – Feb 2025
	Led teams in AI for Operational Research (AIOps), AI for Software, Conversational AI, Time-Series Anomaly Detection, Uncertainty Estimation, and Data Hardness Evaluation	
Senior Manager	Salesforce Research	Palo Alto, CA Nov 2019 – Nov 2021
	Deep learning theory and applications, reinforcement learning, multi-task learning, language modeling, multilingual NER, knowledge graph	
Senior Research Scientist	Salesforce Research	Palo Alto, CA 2018 – 2019
	Deep learning research in large-scale language modeling and vision-language integration	
Senior Applied Researcher	Microsoft AI Research	Sunnyvale, CA 2015 – 2018
	- Developed deep learning systems for recommendation, ranking, and intelligent Q&A - Improved web and local search relevance using neural embeddings	
Research Scientist	Yahoo! Labs	New York, NY 2013 – 2015
	- Designed large-scale ML algorithms for search ads prediction and account security prediction - Leveraged Hadoop, Spark, and Storm for data processing	
Adjunct Professor	New York University	New York, NY 2014
	Taught the "Machine Learning" class.	
Adjunct Professor	Baruch College	New York, NY 2014
	Taught the "Algorithm Design" class.	
Research Intern	Microsoft Research	Redmond, WA 2011 – 2011
	Projects on anomaly detection and time series modeling	

Honors and Awards

- Best Paper Award, Conference on Learning Theory (COLT), 2012
- Award of Excellence – Stars of Tomorrow, Microsoft Research, Asia
- Bachelor's Degree with Honors, Zhejiang University

Representative Publications

Full publication list: [Google Scholar](#)

Large Language Models (LLMs)

- [APIGen-MT: Agentic Pipeline for Multi-Turn Data Generation via Simulated Agent-Human Interplay](#). [NeurIPS Datasets & Benchmarks Track](#), 2025. (*co-corresponding author*). [\[Data\]](#), [\[Model\]](#)
- [APIGen: Automated Pipeline for Generating Verifiable and Diverse Function-Calling Datasets](#). [NeurIPS](#), 2024. [\[Data\]](#), [\[Model\]](#)
- [xLAM: A Family of Large Action Models to Empower AI Agent Systems](#). [NAACL](#), 2024. (*co-corresponding author*). [\[Code\]](#)
- [Retroformer: Retrospective Large Language Agents with Policy Gradient Optimization](#). [ICLR](#), 2024. (*co-corresponding author*).
- [CodeGen: An Open Large Language Model for Code with Multi-Turn Program Synthesis](#). [ICLR](#), 2023. [\[Code\]](#)

AI Agents and Multi-Agent Systems

- [BOLAA: Benchmarking and Orchestrating LLM-Augmented Autonomous Agents](#). [arXiv](#), 2023. [\[Code\]](#)
- [AgentLite: A Lightweight Library for Building and Advancing Task-Oriented LLM Agent System](#). [arXiv](#), 2024. [\[Code\]](#)
- [CRMarena: Understanding the Capacity of LLM Agents to Perform Professional CRM Tasks in Realistic Environments](#). [NAACL](#), 2025. [\[Code\]](#)
- [ActionStudio: A Lightweight Framework for Data and Training of Large Action Models](#). [EMNLP](#), 2025.
- [UserRL: Training Interactive User-Centric Agent via Reinforcement Learning](#). [arXiv](#), 2025. [\[Code\]](#)
- [Enterprise Deep Research: Steerable Multi-Agent Deep Research for Enterprise Analytics](#). [arXiv](#), 2025.
- [MCPEval: Automatic MCP-based Deep Evaluation for AI Agent Models](#). [arXiv](#), 2025. [\[Code\]](#)
- [REX: Rapid Exploration and eXploitation for AI Agents](#). [arXiv](#), 2023.

LLM Reasoning, Routing, and Software Engineering

- [Language Models are Hidden Reasoners: Unlocking Latent Reasoning Capabilities via Self-Rewarding](#). [arXiv](#), 2024.
- [PRACT: Optimizing Principled Reasoning and Acting of LLM Agent](#). [CoNLL](#), 2024.
- [LATTE: Learning to Think with Vision Specialists](#). [arXiv](#), 2024.
- [ToolLibGen: Scalable Automatic Tool Creation and Aggregation for LLM Reasoning](#). [arXiv](#), 2025.
- [xRouter: Training Cost-Aware LLMs Orchestration System via Reinforcement Learning](#). [arXiv](#), 2025.
- [LoCoBench: A Benchmark for Long-Context Large Language Models in Complex Software Engineering](#). [arXiv](#), 2025. [\[Code\]](#)

Theory of Deep Learning and In-Context Learning

- [Transformers as Statisticians: Provable In-Context Learning with In-Context Algorithm Selection](#). [NeurIPS](#), 2023.
- [How do Transformers Learn In-Context Beyond Simple Functions? A Case Study on Learning with Representations](#). [ICLR](#), 2024.
- [How Important is the Train-Validation Split in Meta-Learning?](#) [ICML](#), 2021.
- [Catastrophic Fisher Explosion: Early Phase Fisher Matrix Impacts Generalization](#). [ICML](#), 2021.

- [Ensemble of Averages: Improving Model Selection and Boosting Performance in Domain Generalization](#). [🔗](#) NeurIPS, 2022.

Reinforcement Learning

- [Policy Finetuning: Bridging Sample-Efficient Offline and Online Reinforcement Learning](#). [🔗](#) NeurIPS, 2021.
- [Sample-Efficient Learning of Stackelberg Equilibria in General-Sum Games](#). [🔗](#) NeurIPS, 2021.
- [WarpDrive: Extremely Fast End-to-End Deep Multi-Agent Reinforcement Learning on a GPU](#). [🔗](#) arXiv, 2021.
- [On the Generalization Gap in Reparameterizable Reinforcement Learning](#). [🔗](#) ICML, 2019.

Uncertainty Estimation & Reliability

- [Improved Online Conformal Prediction via Strongly Adaptive Online Learning](#). [🔗](#) ICML, 2023.
- [Understanding the Under-Coverage Bias in Uncertainty Estimation](#). [🔗](#) NeurIPS, 2021.
- [Don't Just Blame Over-parametrization for Over-confidence: Theoretical Analysis of Calibration in Binary Classification](#). [🔗](#) ICML, 2021.

Time Series, Causality and Neural Editing

- [Merlion: End-to-End Machine Learning for Time Series](#). [🔗](#) JMLR, 2023.
- [Causal Layering via Conditional Entropy](#). [🔗](#) PMLR, 2024.
- [Salesforce CausalAI Library: A Fast and Scalable Framework for Causal Analysis of Time Series and Tabular Data](#). [🔗](#) arXiv, 2023.
- [On the Unlikelihood of D-Separation](#). [🔗](#) arXiv, 2023.
- [Editing Arbitrary Propositions in LLMs without Subject Label](#). [🔗](#) arXiv, 2024.

Earlier Work in Machine Learning and Sparse Representation

- [Exact Recovery of Sparsely-Used Dictionaries](#). [🔗](#) COLT, 2012. **Best Paper Award**.
- [Adaptive Dropout with Rademacher Complexity Regularization](#). [🔗](#) ICLR, 2018.
- [Trace ratio vs. ratio trace for dimensionality reduction](#). [🔗](#) CVPR, 2007.

Open Source Projects

Large Action Models and AI Agents

- [xLAM & ActionStudio](#) [🔗](#): A family of large action models for AI agent systems, plus a lightweight framework for action-model data and training.
- [AgentLite](#) [🔗](#): A lightweight library for building task-oriented LLM agent systems.
- [BOLAA](#) [🔗](#): Benchmarking and orchestrating LLM-augmented autonomous agents.
- [CRMArena](#) [🔗](#): A benchmark for evaluating LLM agents on professional CRM tasks.
- [MCPEval](#) [🔗](#): Automatic MCP-based deep evaluation for AI agent models.
- [Enterprise Deep Research](#) [🔗](#): Steerable multi-agent deep research for enterprise analytics.
- [PersonaBench](#) [🔗](#): A benchmark for persona-based conversation systems.
- [MobileAIBench](#) [🔗](#): A benchmark for evaluating AI systems on mobile devices.

Function Calling and API Generation

- [APIGen](#) [🔗](#): Automated pipeline for generating verifiable and diverse function-calling datasets.
- [APIGen-MT](#) [🔗](#): Agentic pipeline for multi-turn data generation via simulated agent-human interplay.

Code Generation and Software Engineering

- [CodeGen](#) [🔗](#): A family of open-source models for code generation.
- [CoDA](#) [🔗](#): Coding language models via diffusion adaptation.
- [LoCoBench](#) [🔗](#): A benchmark for long-context large language models in complex software engineering.
- [Diversity Empowers Intelligence](#) [🔗](#): Integrating expertise of software engineering agents.

Vision and Multimodal Models

- [LATTE](#) [🔗](#): Learning to think with vision specialists.

- [xGen-MM \(BLIP3\)](#) [🔗](#): Multimodal generative models. [\[Code\]](#) [🔗](#)
- [UniControl](#) [🔗](#): A unified diffusion model for controllable image generation.
- [Hive](#) [🔗](#): Harnessing Human Feedback for Instructional Visual Editing.

Reinforcement Learning and Optimization

- [WarpDrive](#) [🔗](#): Extremely fast end-to-end deep multi-agent reinforcement learning on GPU.
- [UserRL](#) [🔗](#): User-centric reinforcement learning for interactive LLM agents.
- [UserBench](#) [🔗](#): An interactive gym environment for user-centric agents.
- [xRouter](#) [🔗](#): Training cost-aware LLM orchestration systems via reinforcement learning.

Prompt Optimization

- [Promptomatix](#) [🔗](#): A powerful framework for LLM prompt optimization.
- [Retroformer](#) [🔗](#): Retrospective large language agents with policy gradient optimization.

Time Series and Causal Learning

- [Merlion](#) [🔗](#): An easy-to-use library for time series anomaly detection and forecasting. [\[Blog\]](#) [🔗](#)
- [CausalAI](#) [🔗](#): A library for causal inference and causal discovery.

Dialog and Conversation Systems

- [Converse](#) [🔗](#): A framework for conversational AI applications.
- [DialogStudio](#) [🔗](#): A comprehensive dialogue understanding benchmark and toolkit.

Professional Service

- Reviewer: NeurIPS, ICML, ICLR, ACL, EMNLP, NAACL, AAAI, JMLR, TPAMI.

Skills

Technical: Machine Learning, Deep Learning, Reinforcement Learning, NLP, Computer Vision, Algorithms, AI Agents, Data Mining

Programming: Python

Frameworks & Tools: PyTorch, TensorFlow, Spark, Hadoop

Languages: English, Chinese